

EarComp 2023: Fourth International Workshop on Earable Computing

Andrea Ferlini

andrea.ferlini@nokia-bell-labs.com
Nokia Bell Labs
Cambridge, UK

Nirupam Roy

niruroy@umd.edu,
University of Maryland, College Park
College Park, MD, USA

Alessandro Montanari

alessandro.montanari@nokia-bell-labs.com
Nokia Bell Labs
Cambridge, UK

Katayoun Farrahi

k.farrahi@soton.ac.uk
University of Southampton
Southampton, UK

ABSTRACT

The objective of the 4th ACM International Workshop on Earable Computing (EarComp 2023) is to provide an academic forum and bring together researchers, practitioners, and design experts to discuss how sensory earables technologies have and can complement human sensing research. It also aims to provide a launchpad for bold and visionary ideas and serve as a catalyst for advancements in this emerging new Earable Computing research space.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices; Ubiquitous and mobile devices**; • **Computing methodologies** → **Machine learning algorithms**.

KEYWORDS

Earable; Human Sensing; Behaviour Analysis; Data Systems

ACM Reference Format:

Andrea Ferlini, Alessandro Montanari, Nirupam Roy, and Katayoun Farrahi. 2023. EarComp 2023: Fourth International Workshop on Earable Computing. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3594739.3605112>

1 BACKGROUND

Sensory earables are increasingly becoming a mainstream computing platform with a promise to fundamentally transform personal-scale human sensing applications. Over the past few years, a number of research efforts in the ubiquitous computing domain have sought to achieve useful, engaging, and sometimes ambitious behavioural analytics with sensory earables including studies of the human

face; of emotion and stress; continuous monitoring of cardiovascular function; oxygen consumption and blood flow; and tracking eating episodes as well as dietary and swallowing activities. At the same time, we have started seeing commercial efforts such as Apple AirPods, Bragi's The Dash, Bose SoundSport, Jabra Elite Sport, and Sony Xperia offering music experience augmented with sensory services including fitness tracking, real-time translations and conversational agents. Naturally, earables are becoming an intense interdisciplinary area of study with many and diverse applications including HCI, empathic communication, behavioural science, health and wellbeing, entertainment, education, and security.

However, as of today, earable computing lacks an academic forum to bring together researchers, practitioners, and design experts from academia and industry to discuss, share, and shape this exciting new area of research. To this end, we want to organise the fourth edition of the ACM Workshop on Earable Computing with a hope that this workshop will serve as a catalyst for advancements in sensory earable technology as well as present a clear sense of direction for the research community to proceed in this space.

As a launchpad, in 2018, we leveraged the Open Earable Platform, eSense (<http://www.esense.io>), from Nokia Bell Labs. These devices are being shared with 70+ academic institutions to accelerate the research in this space. In particular, we expect the workshop will include participants from these institutions reporting on their research with sensory earables to bootstrap this community. For the fourth edition of the workshop, we will also consider research contributions which do not build on top of eSense but present innovative developments with other earable devices.

2 ESENSE AND EARABLE COMPUTING COMMUNITY

Nokia Bell Labs Cambridge developed an open earable computing platform called eSense in 2019. eSense provides a high-definition stereo earable equipped with a microphone, a 6-axis inertial measurement unit, and dual-mode Bluetooth. It allows recording of real-time data streams of three modalities: audio, motion, and proximity. The platform (hardware and associated software libraries) is being widely shared with 70+ academic groups active in the areas of ubicomp, mobile and wearable systems research. It is expected that these groups will contribute to building a community around earable computing by sharing:

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico
© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0200-6/23/10...\$15.00
<https://doi.org/10.1145/3594739.3605112>

- curated and labelled datasets collected with eSense across a variety of projects; and
- models using these datasets.

The eSense portal (<http://www.esense.io>) will be the central point of access for this community, and will host project descriptions, links to data and models, relevant publications of different research groups as well as the EarComp workshop page. We hope, through this portal, academic research groups working on this topic will develop an active community space.

3 SCOPE, SUBMISSIONS, STRUCTURE AND SESSIONS

3.1 Call for Papers

We will solicit three categories of papers. Full papers (up to 6 pages including references) should report a reasonably mature work with earables, and is expected to demonstrate concrete and reproducible results albeit scale may be limited. Experience papers (up to 4 pages including references) that present extensive experiences with implementation, deployment, and operations of earable-based systems. Desirable papers are expected to contain real data as well as descriptions of the practical lessons learned. Short papers (up to 2 pages including references) are encouraged to report novel, and creative ideas that are yet to produce concrete research results but are at a stage where community feedback would be useful. Moreover, we will have a special submission category - "Dataset Paper" - soliciting a 1-2 page document describing a well curated and labeled dataset collected with earables (eventually accompanied by the dataset). Full research papers will be in ACM sigconf template with 2 columns and the accepted papers will be included in the ACM Digital Library. All papers will be digitally available through the workshop website, and the UbiComp adjunct proceedings. For each category of papers, we will offer a "Best Paper" and "Best Dataset" awards sponsored by Nokia Bell Labs. In addition, depending on the quality and depth of the submissions we might consider producing a Book on "Earable Computing" contributed by the authors of the papers, and edited by the Workshop Organizers.

Topics of interest:

- Acoustic Sensing with Earables
- Kinetic Sensing with Earables
- Multi-Modal Learning with Earables
- Multi-Task Learning with Earables
- Active Learning with Earables
- Low-Power Sensing Systems for Earables
- Authentication & Trust mechanisms for Earables
- Quality-Aware Data Collection with Earables
- Experience Sampling with Earables
- Crowd Sourcing with Earables
- Novel UI and UX for Earables
- Auditory Augmented Reality Application with Earables
- Lightweight Deep Learning on Earables
- Tiny Machine Learning on Earables
- Health and Wellbeing Applications of Earables
- Emerging applications of Earables

3.2 Workshop Program

In the third edition of the workshop, we accepted eleven contributions from academic research groups, covering a broad area of research topic related to earable computing.

The 1-day workshop program is intended to be an interactive one and will include two keynotes, at least two technical sessions, and a focused panel session. We plan to invite luminaries in the area of mobile and wearable systems research as keynote speakers to set the tone of the workshop. This will be followed by technical sessions around ten papers, each focusing on specific themes related to Earable Computing:

- (1) Systems for Earables
- (2) Learning Algorithms for Earables
- (3) Novel Lifestyle Applications with Earables

Each session will host several long talks (15 minutes). We will close the workshop by organising a 4-member panel session in which we will bring together some of the key researchers from the academia and industry to discuss the grand challenges for earables shaped by the lessons from their success (and failures) with computing on small devices. The panel will also strive to define 2-3 key challenges that the earable computing community at large should address before the next edition of the workshop.

We intend to keep the workshop open for all UbiComp 2023 attendees. However, we might need to limit attendees depending on the number of submitted research papers.

3.3 Organisation

- Website: Earable Computing Workshop will be hosted at <https://www.esense.io/earcomp2023/index.html>
- Publicity: The workshop will already have visibility through the eSense portal, and the participating research groups. In addition, we will maintain active community engagement through twitter and mailing lists.
- Paper Management: A PCS portal will be setup to manage the submissions.

4 EARCOMP HISTORY

The current proposal is towards organizing the fourth edition of the EarComp workshop and looks forward to holding its tradition of being co-located with UbiComp/ISWC. In the first edition of the workshop, we accepted ten contributions from academic research groups that are working with the eSense computing platform. Following the success of EarComp 2019, in 2021 we run the second edition of EarComp and we accepted nine high quality papers. These contributions covered a broad area of research topics related to earable computing including kinetic sensing, quality-aware data collection and machine learning, data platforms and auditory augmented reality, as well as health and wellbeing applications. Finally, in 2022, we hosted the third edition of the workshop (in Cambridge c/o UbiComp/ISWC 2022, in person) featuring eleven key contributions to the earable computing community. During the past editions, the workshop enumerated technical papers from leading researchers in the field working on earables including KIT, Georgia Tech, Cornell University, University of Cambridge, Google Brain, Apple, and Samsung among others. The workshop also had a compelling keynote talk from key figures both from academia,



Figure 1: EarComp 2019

such as Prof. Romit Roy Choudury (UIUC) and Dr. Jamie A Ward (Goldsmiths University of London), as well as industry leaders like Hannu Kinnunen (co-founder of Oura) and Andrew Bellavia (ex Bragi, Doppler Labs, Amazon, Knowles Corp and founder of Aura-Futurity). Figure 1 shows the photos of the first edition of EarComp in 2019 (in person). More photos can be found at the [link](#).

5 STEERING COMMITTEE MEMBERS

- Michael Beigl, Karlsruhe Institute of Technology
- Fahim Kawsar, Nokia Bell Labs, Cambridge
- Cecilia Mascolo, University of Cambridge

6 PROGRAM COMMITTEE MEMBERS

The program committee members comprise a set of eminent experts in diverse fields relating to earable systems. The names of the program committee members are as follows:

- Xiaoran "Van" Fan, Google
- Dong Ma, Singapore Management University
- Marios Costantinides, Nokia Bell Labs Cambridge
- Longfei Shangguan, University of Pittsburgh
- Yang Liu, University of Cambridge
- Yang Liu, Nokia Bell Labs Cambridge
- Ting Dang, Nokia Bell Labs Cambridge
- Khaldoon Al-Naimi, Nokia Bell Labs Cambridge
- Ashok Thangarajan, Nokia Bell Labs Cambridge
- Shijia Pan, UC Merced
- Jagmohan Chauhan, University of Southampton
- Jay Prakash, Silence Laboratories, Singapore

- Jun Han, Yonsei University
- Wen Hu, UNSW Sydney
- Zhenyu Yan, Chinese University of Hong Kong
- Mi Zhang, Michigan State University
- Rajalakshmi Nandakumar, Cornell Tech
- Bashima Islam, Worcester Polytechnic Institute
- VP Nguyen, University of Texas at Arlington/UMass Amherst
- Ho-Hsiang Wu, Bosch, USA

7 WORKSHOP ORGANIZERS

Alessandro Montanari is a Principal Research Scientist and Tech Lead of the Device Forms team at Nokia Bell Labs in the Pervasive Systems Research Department in Cambridge (UK). He holds a PhD in Computer Science from the University of Cambridge. At Bell Labs he leads a team investigating the architectural and algorithmic challenges of building practical wearable systems for human sensing exploiting ultra-low-power machine learning techniques. His research interests include expanding and improving the capabilities of modern sensory systems for efficient human sensing. He is particularly interested in ultra-wearable and ultra-low-power devices with a focus on novel sensing and computational techniques.

Kate Farrahi is an Assistant Professor in the Vision, Learning and Control group at the University of Southampton. She received her Ph.D. from Ecole Polytechnique Federale de Lausanne (EPFL), Switzerland. Her research is focused on machine learning for health and digital epidemics. Recent research topics include wearables for vitals predictions, contactless (video based) vitals predictions, medical imaging and digital contact tracing via reinforcement learning.

Andrea Ferlini is a Research Scientist at Nokia Bell Labs, Cambridge, UK and holds a PhD in Computer Science from the University of Cambridge, UK. At Bell Labs he is part of the Device Forms team, investigating the potential of earables, as well a number of alternative forms of sensing platforms, for a number of personal-scale applications. His research interests are focused around health and wellbeing sensing. More on his work can be found at <https://andreaferlini.github.io>.

Nirupam Roy is an Assistant Professor in Computer Science at the University of Maryland, College Park (UMD). His research interests are in intelligent acoustics, ambient computing, and embedded

AI. He received his Ph.D. from the University of Illinois, Urbana-Champaign (UIUC), in 2018. His doctoral thesis was selected for the 2019 CSL Ph.D. thesis award at UIUC. Prof. Roy is the recipient of the Valkenburg graduate research award, the Lalit Bahl fellowship, and the outstanding thesis awards from both his Bachelor's and Master's institutes. His research received multiple best paper and demonstration awards (MobiSys - 2017, 2021, 2022; SenSys - 2020) and was selected for the ACM SIGMOBILE research highlights. He received the NSF CAREER award in 2023. His research projects have been featured in news media such as the MIT Technology Review, The Telegraph, and The Huffington Post. At UMD, he directs the iCoSMoS lab which has been supported by NSF, Meta Research, Maryland Transportation Institute, USG, and Saab, inc.